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E-Paper API Analysis

Raspberry Pi and Jetson Nano have their own programs. As they are both embedded systems, their compatibilities are high.

Examples contain three parts, hardware interface, SPI driver and the GUI functions.

1.

Hardware Interface

We have carried out the underlying package, the hardware platform is different, the internal is similar. If you want to know the internal, you can refer to the corresponding documents.

You can see many definitions in DEV_Config.py(Jetson Nano) or DEV_Config.h(Raspberry Pi), dev_hardware_SPI.py(Jetson Nano) or dev_hardware_SPI.h(Raspberry Pi), and dev_config.h(Pico). The RASPBERRY library is used by default. If you want to use the Wiringpi library, you can open Raspberry Pi, dev_config.h and RaspberryPi.mk(Raspberry and Jetson Nano) to make changes.

```
# Data type
#define UBYTE  uint8_t
#define UWORD  uint16_t
#define UDOUBLE uint32_t

# set and Get
void DEV_Delay_Ms(UWORD);
void DEV_Delay_Us(UWORD);
```

Note: The 4-wire and 3-wire functions are used to configure GPIOs. SPI writes data mode (the 4-wire) function is used, and the consumption of a Paper should be 0 in the standby state. The 3-wire function is used:

```
# GPIO Read/Write
void DEV_Digital_Write(UWORD Pin, UBYTE Value);
UBYTE DEV_Digital_Read(UWORD Pin);

# SPI transmits data.
void DEV_SPI_WriteByte(UBYTE Value);
```

SPI Driver

The driver files is saved under Raspberry/Python/examples/lib/waveshare

epd1in54.py	epd2in9.py	epd2in9bc.py	epd2in9d.py
epd1in54b.py	epd2in13.py	epd2in13bc.py	epd2in13d.py
epd1in54c.py	epd2in7.py	epd2in7b.py	epd2in9_V2.py
epd2in13_V2.py	epd2in13_V3.py	epd2in13_V4.py	epd2in7_V2.py
epd4in2.py	epd4in2b_V2.py	epd5in83.py	epd5in83b_V2.py
epd5in83bc.py	epd5in65f.py	epd5in83b_V3.py	epd5in83b_V4.py
epd7in3f.py	epd7in3g.py	epd7in5.py	epd7in5b_V2.py
epd7in5bc.py	epd7in5_HD.py	epd7in5_V2.py	epd7in5_V3.py
epd7in5_V4.py	epd7in5b_HD.py	epd7in5d.py	epd7in5h.py
epd13in3k.py	epd13in3E.py	epd13in3f.py	epd13in3g.py

Open it to see the following functions

```
# SSD1681initialize: please use it when the EPD starts to work or exits the sleep mode.
def init(self):
    if(epd.Init() == 0):
        epd.SetWindows(0, 0, epd.width, epd.height)
        epd.SetCursor(0, 0)
        epd.ClearFrame()
    else:
        exit()

# TP2144initialize: please use it when the EPD starts to work or exits the sleep mode.
def init(self):
    if(epd.Init() == 0):
        epd.ClearFrame();
    else:
        exit()
```

Note that the initialization of the screen models 4.2 to 7.5 (b), there is EPD_2IN9D_Init(), 2.9 to 7.5b, is EPD_7IN5_V2_Init(). Although the display colors are different, the driver code of 7.5B and 7.5C is the same.

- Transfer a frame of picture data and open the display.

```
    epd.display(epd.getbuffer(Himage))
    epd.DisplayPartial(epd.getbuffer(Himage))
```

Note that the following are special cases:

```
    epd.TP_Display(epd.getbuffer(Himage))
    epd.EPD_7IN5_V2_Display(epd.getbuffer(Himage))
    epd.EPD_7IN5_V2_DisplayPart(epd.getbuffer(Himage))
    epd.EPD_7IN5_V2_DisplayPart4Gray(epd.getbuffer(Himage))
```

- Enter sleep mode

```
    epd.sleep()
```

Note that after entering sleep mode, there are only two ways to work again: the first is a hardware reset, and the second is to call the initialization function again. Where our repository the six screen models 4.2 to 7.5B, there is EPD_2IN9D_Init(), 2.9 to 7.5B is EPD_7IN5B_V2_Init(). Although the display colors are different, the driver code of 7.5B and 7.5C is the same.

GUI Functions

GUI files can be found in Raspberry/Python/examples/GUI/GUI_Paint.py(Jetson Nano)

gui.py.conf	drawtext.py.conf	1in54
gui.py.conf1	drawtext2.py.conf	2in9
bitmap	drawtext_chinese.py.conf	4in2
bitmap1	drawtext_time.py.conf	7in5

The fonts can be found in the Raspberry/Pi & Jetson/font2 (16Fonts directory

Font0.ttc	DejaVuSans.ttf	1.5KB	2.0KB
Font1.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
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Create an Image Buffer

```
    epd.Init_GetWindowBuff(mode, image, BWRB\_Buff, Width, Height, BWRB\_Color)
```

- Image: the image buffer
- BWRB_Buff: the buffer
- Height: Height of the image
- Width: Width of the image
- Color: Color of the image

Select Image Buffer

```
    epd.Init_SelectImageBuff(image)
```

- The image buffer is a portion of the image buffer's first address.

Rotate Image

This function should be used after Paint_SelectImage()

```
    epd.Init_SetRotate(ROTATE, Mirror)
```

- Rotate: The angle rotated. It should be ROTATE_0, ROTATE_90, ROTATE_180, ROTATE_270.
- Mirror: For different orientations, the position of the first point is different. Here we take Rotate as an example.

Mirror

```
    epd.Init_SetMirror(mirror)
```

- mirror: The type of mirroring. BARREL_MIRROR, MIRROR_HORIZONTAL, MIRROR_VERTICAL, MIRROR_ORIGIN

Set Pixel

This function is used to set the position and color of the pixel

```
    epd.Init_SetPixel(X\_Point, Y\_Point, Color, Mode=None)
```

- Xpoint: The x axis coordination of pixel
- Ypoint: The y axis coordination of pixel
- Color: The color of the pixel

Clear

This function is used to clear the e-Paper

```
    epd.Init_Clear(Color)
```

- Color: The color of the display

Clear window

This function is used to clear a partial area

```
    epd.Init_ClearWindows(XStart, YStart, XEnd, YEnd, Color)
```

- Xstart: The x axis coordination of the start point
- Ystart: The y axis coordination of the start point
- Xend: The x axis coordination of the end point
- Yend: The y axis coordination of the end point
- Color: The color of the windows

Draw point

This function is used to draw points.

```
    epd.Init_DrawPoint(XPoint, YPoint, Color, DotStyle=DOT_PIXEL_1X1)
```

- Xpoint: The x axis coordination of the point
- Ypoint: The y axis coordination of the point
- Color: The color of the point
- DotStyle: The width of dotted. It does not available.

Draw line

```
    epd.Init_DrawLine(XStart, YStart, XEnd, YEnd, Color, LINE\_STYLE\_SOLID, Line\_Style)
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
- Line_width: The width of display. It does not available.

Draw rectangle

```
    epd.Init_DrawRectangle(XStart, YStart, XEnd, YEnd, Color, DRAW\_FILL\_EMPTY, Draw\_Fill)
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
- Line_width: The width of display. It does not available.

Draw circle

```
    epd.Init_DrawCircle(X\_Center, Y\_Center, Radius, Color, DRAW\_FILL\_EMPTY, Draw\_Fill)
```

- X_Center: X coordinate of the center of the circle
- Y_Center: Y coordinate of the center of the circle
- Radius: The radius of the circle
- Color: The color of the circle
- Line_width: The width of the arc. It does not available. By default width

Draw arc

```
    epd.Init_DrawArc(X\_Center, Y\_Center, Radius, Color, LINE\_STYLE\_SOLID, Line\_Style)
```

- X_Center: X coordinate of the center of the arc
- Y_Center: Y coordinate of the center of the arc
- Radius: The radius of the circle
- Color: The color of the arc
- Line_width: The width of the arc. It does not available.

Draw string

```
    epd.Init_DrawString_EN(XStart, YStart, pChar, pFont, Color_Foreground, Color_Background)
```

Parameter:

- Xstart: X coordinate of the left top pixel of character.
- Ystart: Y coordinate of the left top pixel of character.
- pChar: The character.
- Font: It fonts are available:
  - Font12: 7*12
  - Font16: 11*16
  - Font20: 14*20
  - Font24: 17*24
- Color_Foreground: the color of character.
- Color_Background: the color of background.

Draw string

Use pen Draw() Function to draw the text and draw a string.

```
    epd.Init_DrawString_EN(XStart, YStart, text, pFont, Color_Foreground, Color_Background)
```

Parameter:

- Xstart: X coordinate of left top pixel of characters.
- Ystart: Y coordinate of left top pixel of characters.
- pString: Pointer of string.
- Font: It fonts are available:
  - Font8: 5*8
  - Font12: 7*12
  - Font16: 11*16
  - Font20: 14*20
  - Font24: 17*24
- Color_Foreground: the color of the fonts.
- Color_Background: the color of the background.

Draw Image

Send image data of BMP file to buffer

```
    epd.Init_DrawBitMap(color) #color=0x00 or 0xff
```

Parameter:

- Image: buffer the first address of image data in the buffer.
- **Read the local BMP pictures and write it to buffer.**

Linux platforms like Jetson Nano and Raspberry Pi support directly operating BMP pictures. Raspberry Pi & Jetson Nano/Raspberry/Python/examples/GUI/GUI_BMP.py

```
BMPIN_RED(Image,Xstart,Ystart,Xend,Ystart,Xend,Ystart)
```

Parameters:

- path: The path of BMP pictures
- Xstart: X coordinate of the left top of pictures, default 0.
- Ystart: Y coordinate of the left top of pictures, default 0.

Testing Code

The first three chapters introduce the classes Linux three-layer code structure, here is a little explanation of the user test code.

Raspberry Pi and Jetson Nano, in the directory Raspberry/Pi_examples/examples/ for all test codes, multiple details can be made in each. In this directory,

If you need to run the "Timer in paper test program, you need to rename the 42 line sheet.

```
./epd_5in83_test1.
```

to

```
EPD_5in83_test1.
```

Then compile it again and run.

```
make clean
make
./epd. test
```

Python

For Jetson Nano/Raspberry Pi, based on python2.7 and python3.

For python, its code are not as standardized as in Raspberry Pi and Jetson Nano. Raspberry/Pi/python/python/lib/

Bottom Interface

The bottom/top file package the underlying interface.

- Initialize module and set handle:

```
    def module_init()
    def module_exit()
```

Note:

- The functions are used to set GPIO options and other strong o Paper.
- If the board you have is printed and fixed, the module-drive-law-who-made-after-Module_Exit() be are test, the current is about it in this module.

- GPIO Read/Write:

```
    def digital_write(pin, value)
    def digital_read(pin)
```

- SPI Write data:

```
    def spi_writebyte(data)
```

Driver Interface

Initialize Each needs only 1 if it is 1.54inch e-paper, it is epd1in54(), and so on.

- Initialize a paper: this function should be used at the beginning. It can also be used to wake up a Paper from Sleep mode.

```
    for i in range(1,4):exec("epd = epd.epd2in13_V2(partial=False)")
    epd.init(epd.lut_full_update)
    epd.Clear(0xff)
    epd.init(epd.lut_partial_update)
```

- Clear a paper: This function is used to clear a Paper to white.

```
    epd.Clear(0xff)
```

- Sleep: This function is used to sleep a Paper to white.

```
    epd.sleep()
```

Display

Where our repository the six screen models 4.2 to 7.5B, there is EPD_2IN9D_Clear(), 2.9 to 7.5B, is EPD_7IN5_Clear(). Although the display colors are different, the driver code of 7.5B and 7.5C is the same.

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GUI Functions

GUI files can be found in Raspberry/Pi & Jetson/Python2/GUI/GUI_Paint.py

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gui.py.conf1	drawtext2.py.conf	2.0KB
bitmap	drawtext_chinese.py.conf	2.0KB
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Create an Image Buffer

```
    epd.Init_GetWindowBuff(mode, image, BWRB\_Buff, Width, Height, BWRB\_Color)
```

- Image: the image buffer
- BWRB_Buff: the buffer
- Height: Height of the image
- Width: Width of the image
- Color: Color of the image

Select Image Buffer

```
    epd.Init_SelectImageBuff(image)
```

- The image buffer is a portion of the image buffer's first address.

Rotate Image

This function should be used after Paint_SelectImage()

```
    epd.Init_SetRotate(ROTATE, Mirror)
```

- Rotate: The angle rotated. It should be ROTATE_0, ROTATE_90, ROTATE_180, ROTATE_270.
- Mirror: For different orientations, the position of the first point is different. Here we take Rotate as an example.

Mirror

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    epd.Init_SetMirror(mirror)
```

- mirror: The type of mirroring. BARREL_MIRROR, MIRROR_HORIZONTAL, MIRROR_VERTICAL, MIRROR_ORIGIN

Set Pixel

This function is used to set the position and color of the pixel

```
    epd.Init_SetPixel(XPoint, YPoint, Color, Mode=None)
```

- Xpoint: The x axis coordination of pixel
- Ypoint: The y axis coordination of pixel
- Color: The color of the pixel

Clear

This function is used to clear the e-Paper

```
    epd.Init_Clear(Color)
```

- Color: The color of the display

Clear window

This function is used to clear a partial area

```
    epd.Init_ClearWindows(XStart, YStart, XEnd, YEnd, Color)
```

- Xstart: The x axis coordination of the start point
- Ystart: The y axis coordination of the start point
- Xend: The x axis coordination of the end point
- Yend: The y axis coordination of the end point
- Color: The color of the windows

Draw point

This function is used to draw points.

```
    epd.Init_DrawPoint(XPoint, YPoint, Color, DotStyle=DOT_PIXEL_1X1 )
```

- Xpoint: The x axis coordination of the point
- Ypoint: The y axis coordination of the point
- Color: The color of the point
- DotStyle: The width of dotted. It does not available.

Draw line

```
    epd.Init_DrawLine(XStart, YStart, XEnd, YEnd, Color, LINE\_STYLE\_SOLID, Line\_Style )
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
- Line_width: The width of display. It does not available.

Draw rectangle

```
    epd.Init_DrawRectangle(XStart, YStart, XEnd, YEnd, Color, DRAW\_FILL\_EMPTY, Draw\_Fill )
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
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Draw circle

```
    epd.Init_DrawCircle(X\_Center, Y\_Center, Radius, Color, DRAW\_FILL\_EMPTY, Draw\_Fill )
```

- X_Center: X coordinate of the center of the circle
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- Radius: The radius of the circle
- Color: The color of the circle
- Line_width: The width of the arc. It does not available. By default width

Draw arc

```
    epd.Init_DrawArc(X\_Center, Y\_Center, Radius, Color, LINE\_STYLE\_SOLID, Line\_Style )
```

- X_Center: X coordinate of the center of the arc
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- Radius: The radius of the circle
- Color: The color of the arc
- Line_width: The width of the arc. It does not available.

Draw string

```
    epd.Init_DrawString_EN(XStart, YStart, pChar, pFont, Color_Foreground, Color_Background )
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Draw image

Send image data of BMP file to buffer

```
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```

Parameter:

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This function is used to set the position and color of the pixel

```
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```

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- Ypoint: The y axis coordination of pixel
- Color: The color of the pixel

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This function is used to clear the e-Paper

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    epd.Init_Clear(Color)
```

- Color: The color of the display

Clear window

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```
    epd.Init_ClearWindows(XStart, YStart, XEnd, YEnd, Color)
```

- Xstart: The x axis coordination of the start point
- Ystart: The y axis coordination of the start point
- Xend: The x axis coordination of the end point
- Yend: The y axis coordination of the end point
- Color: The color of the windows

Draw point

This function is used to draw points.

```
    epd.Init_DrawPoint(XPoint, YPoint, Color, DotStyle=DOT_PIXEL_1X1 )
```

- Xpoint: The x axis coordination of the point
- Ypoint: The y axis coordination of the point
- Color: The color of the point
- DotStyle: The width of dotted. It does not available.

Draw line

```
    epd.Init_DrawLine(XStart, YStart, XEnd, YEnd, Color, LINE\_STYLE\_SOLID, Line\_Style )
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
- Line_width: The width of display. It does not available.

Draw rectangle

```
    epd.Init_DrawRectangle(XStart, YStart, XEnd, YEnd, Color, DRAW\_FILL\_EMPTY, Draw\_Fill )
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
- Line_width: The width of display. It does not available.

Draw circle

```
    epd.Init_DrawCircle(X\_Center, Y\_Center, Radius, Color, DRAW\_FILL\_EMPTY, Draw\_Fill )
```

- X_Center: X coordinate of the center of the circle
- Y_Center: Y coordinate of the center of the circle
- Radius: The radius of the circle
- Color: The color of the circle
- Line_width: The width of the arc. It does not available. By default width

Draw arc

```
    epd.Init_DrawArc(X\_Center, Y\_Center, Radius, Color, LINE\_STYLE\_SOLID, Line\_Style )
```

- X_Center: X coordinate of the center of the arc
- Y_Center: Y coordinate of the center of the arc
- Radius: The radius of the circle
- Color: The color of the arc
- Line_width: The width of the arc. It does not available.

Draw string

```
    epd.Init_DrawString_EN(XStart, YStart, pChar, pFont, Color_Foreground, Color_Background )
```

Parameter:

- Xstart: X coordinate of left top pixel of characters.
- Ystart: Y coordinate of left top pixel of characters.
- pString: Pointer of string.
- Font: It fonts are available:
  - Font8: 5*8
  - Font12: 7*12
  - Font16: 11*16
  - Font20: 14*20
  - Font24: 17*24
- Color_Foreground: the color of the fonts.
- Color_Background: the color of the background.

Draw image

Send image data of BMP file to buffer

```
    epd.Init_DrawBitMap(color) #color=0x00 or 0xff
```

Parameter:

- Image: buffer the first address of image data in the buffer.
- **Read the local BMP pictures and write it to buffer.**

Linux platforms like Jetson Nano and Raspberry Pi support directly operating BMP pictures. Raspberry Pi & Jetson Nano/Raspberry/Python/examples/GUI/GUI_BMP.py

```
BMPIN_RED(Image,Xstart,Ystart,Xend,Ystart,Xend,Ystart)
```

Parameters:

- path: The path of BMP pictures
- Xstart: X coordinate of the left top of pictures, default 0.
- Ystart: Y coordinate of the left top of pictures, default 0.

Testing Code

The first three chapters introduce the classes Linux three-layer code structure, here is a little explanation of the user test code.

Raspberry Pi and Jetson Nano, in the directory Raspberry/Pi_examples/examples/ for all test codes, multiple details can be made in each. In this directory,

If you need to run the "Timer in paper test program, you need to rename the 42 line sheet.

```
./epd_5in83_test1.
```

to

```
EPD_5in83_test1.
```

Then compile it again and run.

```
make clean
make
./epd. test
```

Python

For Jetson Nano/Raspberry Pi, based on python2.7 and python3.

For python, its code are not as standardized as in Raspberry Pi and Jetson Nano. Raspberry/Pi/python/python/lib/

Bottom Interface

The bottom/top file package the underlying interface.

- Initialize module and set handle:

```
    def module_init()
    def module_exit()
```

Note:

- The functions are used to set GPIO options and other strong o Paper.
- If the board you have is printed and fixed, the module-drive-law-who-made-after-Module_Exit() be are test, the current is about it in this module.

- GPIO Read/Write:

```
    def digital_write(pin, value)
    def digital_read(pin)
```

- SPI Write data:

```
    def spi_writebyte(data)
```

Driver Interface

Initialize Each needs only 1 if it is 1.54inch e-paper, it is epd1in54(), and so on.

- Initialize a paper: this function should be used at the beginning. It can also be used to wake up a Paper from Sleep mode.

```
    for i in range(1,4):exec("epd = epd.epd2in13_V2(partial=False)")
    epd.init(epd.lut_full_update)
    epd.Clear(0xff)
    epd.init(epd.lut_partial_update)
```

- Clear a paper: This function is used to clear a Paper to white.

```
    epd.Clear(0xff)
```

- Sleep: This function is used to sleep a Paper to white.

```
    epd.sleep()
```

Display

Where our repository the six screen models 4.2 to 7.5B, there is EPD_2IN9D_Clear(), 2.9 to 7.5B is EPD_7IN5_Clear(). Although the display colors are different, the driver code of 7.5B and 7.5C is the same.

Note that after entering sleep mode, there are only two ways to work again: the first is a hardware reset, and the second is to call the initialization function again.

Where our repository the six screen models 4.2 to 7.5B, there is EPD_2IN9D_Init(), 2.9 to 7.5B is EPD_7IN5B_V2_Init(). Although the display colors are different, the driver code of 7.5B and 7.5C is the same.

GUI Functions

GUI files can be found in Raspberry/Pi & Jetson/Python3/GUI/GUI_Paint.py

gui.py.conf	drawtext.py.conf	1.5KB
gui.py.conf1	drawtext2.py.conf	2.0KB
bitmap	drawtext_chinese.py.conf	2.0KB
bitmap1	drawtext_time.py.conf	2.0KB

The fonts can be found in the Raspberry/Pi & Jetson/font2 (16Fonts directory

Font0.ttc	DejaVuSans.ttf	1.5KB	2.0KB
Font1.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
Font10.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
Font11.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
Font12.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
Font13.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
Font14.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB
Font15.ttc	DejaVuSansBoldOblique.ttf	2.0KB	2.0KB

Create an Image Buffer

```
    epd.Init_GetWindowBuff(mode, image, BWRB\_Buff, Width, Height, BWRB\_Color)
```

- Image: the image buffer
- BWRB_Buff: the buffer
- Height: Height of the image
- Width: Width of the image
- Color: Color of the image

Select Image Buffer

```
    epd.Init_SelectImageBuff(image)
```

- The image buffer is a portion of the image buffer's first address.

Rotate Image

This function should be used after Paint_SelectImage()

```
    epd.Init_SetRotate(ROTATE, Mirror)
```

- Rotate: The angle rotated. It should be ROTATE_0, ROTATE_90, ROTATE_180, ROTATE_270.
- Mirror: For different orientations, the position of the first point is different. Here we take Rotate as an example.

Mirror

```
    epd.Init_SetMirror(mirror)
```

- mirror: The type of mirroring. BARREL_MIRROR, MIRROR_HORIZONTAL, MIRROR_VERTICAL, MIRROR_ORIGIN

Set Pixel

This function is used to set the position and color of the pixel

```
    epd.Init_SetPixel(XPoint, YPoint, Color, Mode=None)
```

- Xpoint: The x axis coordination of pixel
- Ypoint: The y axis coordination of pixel
- Color: The color of the pixel

Clear

This function is used to clear the e-Paper

```
    epd.Init_Clear(Color)
```

- Color: The color of the display

Clear window

This function is used to clear a partial area

```
    epd.Init_ClearWindows(XStart, YStart, XEnd, YEnd, Color)
```

- Xstart: The x axis coordination of the start point
- Ystart: The y axis coordination of the start point
- Xend: The x axis coordination of the end point
- Yend: The y axis coordination of the end point
- Color: The color of the windows

Draw point

This function is used to draw points.

```
    epd.Init_DrawPoint(XPoint, YPoint, Color, DotStyle=DOT_PIXEL_1X1 )
```

- Xpoint: The x axis coordination of the point
- Ypoint: The y axis coordination of the point
- Color: The color of the point
- DotStyle: The width of dotted. It does not available.

Draw line

```
    epd.Init_DrawLine(XStart, YStart, XEnd, YEnd, Color, LINE\_STYLE\_SOLID, Line\_Style )
```

- Xstart: Start coordinate of X axis of the rectangle
- Ystart: Start coordinate of Y axis of the rectangle
- Xend: End coordinate of X axis of the rectangle
- Yend: End coordinate of Y axis of the rectangle
- Color: The color of the rectangle
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